Methods In Differential Equations

Introduction

Differential equations relate variables and their derivatives. A first-order equation contains a first derivative but no higher derivatives.

Temperature loss proportional to temperature above the surroundings:

$$\frac{dT}{dt} = -kT.$$

Here T is excess temperature.

Population change proportional to $P(1 - P/M)$, with limiting population M (Verhulst–Pearl model):

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right).$$

GDP x grows at a rate proportional to its current value:

$$\frac{dx}{dt} = kx.$$

Applications in physics and engineering include radioactive decay, mixing fluids, cooling materials and bodies falling against resistance.

Separation of variables applies when the right-hand side is a product of one function of x and one of y:

$$\frac{dy}{dx} = f(x)g(y).$$

For $g(y) \neq 0$, divide through:

$$\frac{1}{g(y)}\frac{dy}{dx} = f(x).$$

Multiply by dx and integrate:

$$\int \frac{1}{g(y)}\,dy = \int f(x)\,dx.$$

Include an arbitrary integration constant. Check separately for constant solutions satisfying $g(y) = 0$.

Examples 4

Find the general solution of $dy/dx = 2$.

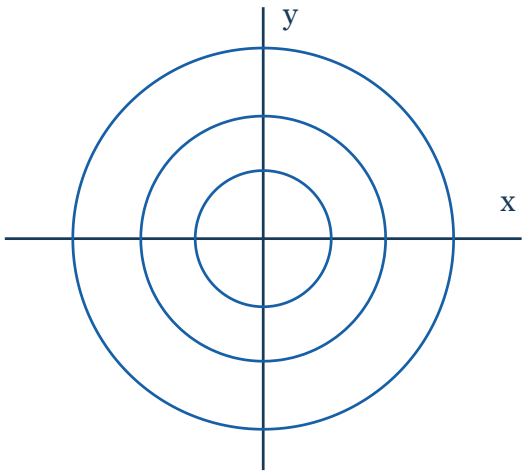
Solution

Why is it called the general solution?

Family

Find the general solution of $dy/dx = -x/y$, where $y \neq 0$.

Solution



Examples 5

Find the general solution of $dy/dx = xy + x$.

Separating

Find the general solution of $dy/dx = -y/x$, $x \neq 0$.

Solution

Using Reverse Product Rule 6

We will solve $y' + Py = Q$, where P, Q are functions of x. First practise reverse product rule.

Find the general solution of $x^3y' + 3x^2y = \sin x$. The variables do not separate. What do you notice about the left-hand side?

Recognise

Differentiate x^2y.

Rewrite $x^4y' + 4x^3y$.

Differentiate $y \sin x$.

Rewrite $e^xy' + e^xy$.

Rewrite $(\ln x)y' + y/x$.

The coefficient of y' becomes the factor multiplying y inside the derivative.

Using Reverse Product Rule 7

Solve $x^3y' + 3x^2y = \sin x$.

Solution

Test: $\frac{1}{x}y' - \frac{y}{x^2} = e^x$.

Test: $4xyy' + 2y^2 = x^2$.

Solving First-order Linear Equations 8

What if reverse product rule is not immediately available? Solve

$y' - 4y = e^x$.

Multiply by the integrating factor $e^{\int P\,dx}$, where $P = -4$.

Integrate

Factor

Multiply

Why the Integrating Factor Works 9

For $y' + Py = Q$, suppose $f(x)$ is an integrating factor. Multiply through:

$$fy' + fPy = fQ.$$

To use reverse product rule, the left side must equal $fy' + f'y$.

Comparing coefficients:

$$f' = fP.$$

Choose $f > 0$, divide and integrate:

$$\int \frac{f'(x)}{f(x)}\,dx = \int P\,dx$$

$$\ln f(x) = \int P\,dx.$$

$$f(x) = e^{\int P\,dx}.$$

A non-zero constant multiplier does not affect the method.

When the Derivative Has a Coefficient 10

Solve $\cos x\, y' + 2y \sin x = \cos^4 x$. What should we do first?

Divide

Multiply

Factor

Integrate

Test Your Understanding 11

Edexcel FP2 (Old), June 2011 Q3

Find the general solution of

$$xy' + 5y = \frac{\ln x}{x}, \quad x > 0.$$

Factor

Solution

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Exercise 7A — Core Pure Book 2, pages 151–153.

Exercise 7A — Core Pure Book 2, pages 151–153.

Exercise 7A — Core Pure Book 2 Pages 151–153 14

Exercise 7A — Core Pure Book 2, pages 151–153.

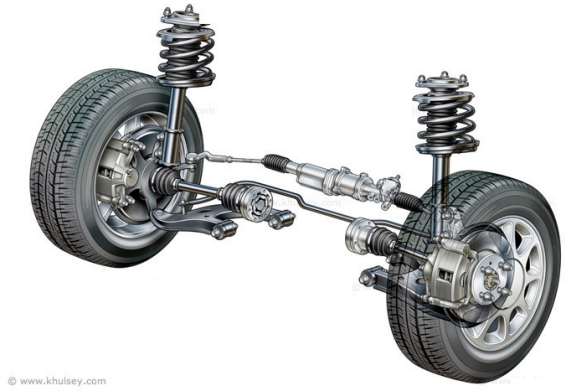
Exercise 7A — Core Pure Book 2, pages 151–153.

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Exercise 7A — Core Pure Book 2, pages 151–153.

Second-order differential equations involve a second derivative.

Car shock absorbers provide an example. Damping opposes velocity; the restoring spring force opposes displacement from equilibrium. Their magnitudes are proportional to velocity and extension respectively.



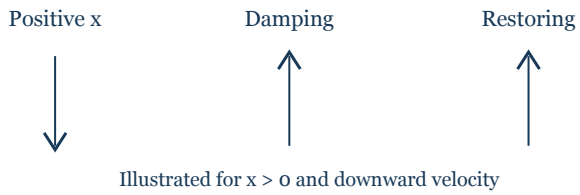
Take downward displacement x as positive, measured from equilibrium. Newton's second law gives:

$$m\frac{d^2x}{dt^2}$$

$$=$$

$$-b\frac{dx}{dt}$$

$$-cx.$$



For $x > 0$ and $\dot{x} > 0$, both forces and the resulting acceleration are upward. Positive coordinate direction does not specify actual acceleration.

For $ay' + by = 0$, the solution is $y = Ae^{-bx/a}$. Try a similar form for

$ay'' + by' + cy = 0, \quad a \neq 0.$

Let $y = Ae^{mx}$.

$$y' = Ame^{mx}, \qquad y'' = Am^2e^{mx}.$$

$$aAm^2e^{mx} + bAme^{mx} + cAe^{mx} = 0$$
$$Ae^{mx}(am^2 + bm + c) = 0.$$

For a non-zero trial solution, $A \neq 0$, so

$am^2 + bm + c = 0.$

This is the auxiliary equation: its roots determine the solutions of the differential equation.

Distinct Real Roots 19

If m is a root of $am^2 + bm + c = 0$, then Ae^{mx} solves $ay'' + by' + cy = 0$.

For distinct real roots α, β:

$y = Ae^{\alpha x} + Be^{\beta x}$.

Here A, B are arbitrary constants.

A zero right-hand side makes the equation homogeneous.

Solve $2y'' + 5y' + 3y = 0$.

Solution

Repeated Roots 20

For two equal roots α, where $b^2 - 4ac = 0$:

$y = (A + Bx)e^{\alpha x}$.

Show that $(A + Bx)e^{3x}$ satisfies

$y'' - 6y' + 9y = 0$.

The auxiliary equation is $(m - 3)^2 = 0$.

Derivatives

Verify

Why not $Ae^{\alpha x} + Be^{\alpha x}$? It combines to just one arbitrary constant. The extra x gives an independent solution, analogous to the distinct terms $A/x + B/x^2$ used for repeated partial-fraction factors.

Pure Imaginary Roots 21

Complex roots work similarly to distinct real roots.
Solve $y'' + 16y = 0$.

Roots

Rewrite

For roots $\pm i\omega$:

$y = A\cos\omega x + B\sin\omega x.$

General Complex Roots 22

What about general complex roots? Solve
$y'' - 6y' + 34y = 0$.

Roots

Rewrite

For roots $p \pm iq$:

$y = e^{px}(A\cos qx + B\sin qx),$

with real arbitrary constants A, B.

Quickfire Questions 23

Find general solutions for these auxiliary equations and supplied roots.

$m^2 + 6m + 8 = 0$, roots $-2, -4$

$m^2 - 1 = 0$, roots ± 1

$m^2 - 2m + 1 = 0$, roots 1

$m^2 + 4 = 0$, roots $\pm 2i$

$m^2 + 10m + 25 = 0$, roots -5

$m^2 - 12m + 45 = 0$, roots $6 \pm 3i$

$m^2 + 10 = 0$, roots $\pm i\sqrt{10}$

6

$m^2 + 2m + 5 = 0$, roots $-1 \pm 2i$

Exercise 7B — Core Pure Book 2 Pages 156–157 24

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Exercise 7B — Core Pure Book 2, pages 156–157.

Particular Integrals 29

What if the right-hand side is a function of x?

$ay'' + by' + cy = f(x)$.

This is non-homogeneous when f is not identically zero.

First solve the homogeneous equation:

$ay'' + by' + cy = 0$.

Its general solution is the complementary function (CF).

Next find one solution of

$ay'' + by' + cy = f(x)$

by choosing an appropriate trial function and comparing coefficients. This is the particular integral (PI).

$y = \text{CF} + \text{PI}$.

The differential operator gives zero for the CF and $f(x)$ for the PI; adding gives $f(x)$.

Constant Right-hand Side 30

Find a particular integral of

$y'' - 5y' + 6y = 3$.

Try a constant $y = \lambda$.

First derivative

Second derivative

Substitute

Hence find the general solution.

General solution

Linear Right-hand Side 31

Solve $y'' - 5y' + 6y = 2x$. The forcing is linear, so try

Trial

First derivative

Second derivative

Constant

Linear

Solution

Substitute:

$(6\mu - 5\lambda) + 6\lambda x = 2x.$

Quadratic Right-hand Side 32

Solve $y'' - 5y' + 6y = 3x^2$. Try a quadratic:

Trial

First derivative

Second derivative

Solution

Exponential and Trigonometric Forcing 33

Solve $y'' - 5y' + 6y = e^x$.

Trial 1

Derivative 1

Derivative 2

Solution 1

Solve $y'' - 5y' + 6y = 13\sin 3x$.

Trial 2

Derivative 3

Derivative 4

Solution 2

When the Trial Appears in the CF 34

A trial PI cannot already belong to the CF. This is like needing independent solutions for repeated auxiliary roots.

Solve $y'' - 5y' + 6y = e^{2x}$.

The CF is $Ae^{2x} + Be^{3x}$. What goes wrong if we try $y = \lambda e^{2x}$?

Failure

Instead try

New trial

Coefficient

Solution

A Constant Term in the CF 35

The PI must be independent of the CF. Solve $y'' - 2y' = 3$.

Roots

CF

A constant trial is already in the CF. Instead use

PI

Solution

Multiply a duplicated trial by x until it is independent of the CF. A repeated root may require x^2, rather than only one extra x.

Summary So Far 36

Right-hand side	Trial particular integral
k	λ
$ax + b$	$\lambda + \mu x$
$ax^2 + bx + c$	$\lambda + \mu x + \nu x^2$
ke^{px}	λe^{px}
$m\cos\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$
$m\sin\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$
$m\cos\omega x + n\sin\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$

Modify a trial which duplicates a CF term by multiplying by a sufficient power of x. An exam question may suggest the modified trial.

Test Your Understanding 37

June 2010 Q8

(a) Find λ for which $y = \lambda x \sin 5x$ is a PI of $y'' + 25y = 3\cos 5x$.

(4 marks)

Notice the extra factor x.

a

(b) Hence find the general solution. (3 marks)

b

Test Your Understanding 38

June 2012 Q4

Find the general solution of

$$\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 2\cos t - \sin t.$$

Here x is the dependent variable and t the independent variable.

Solution

Exercise 7C — Core Pure Book 2, pages 161–162.

Exercise 7C — Core Pure Book 2 Pages 161–162 40

Exercise 7C — Core Pure Book 2, pages 161–162.

Exercise 7C — Core Pure Book 2, pages 161–162.

Exercise 7C — Core Pure Book 2 Pages 161–162 42

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Boundary Conditions 44

Conditions let us determine arbitrary constants. Find y given

$y'' - y = 2e^{x}, \qquad y(0) = y'(0) = 0.$

General solution

Use y(0)

Differentiate

Use derivative

Constants

Solution

Test Your Understanding 45

June 2010 Q8 revisited

$y'' + 25y = 3\cos 5x$.

(c) Given $y(0) = 0$, $y'(0) = 5$, find $y = f(x)$. (5 marks)

Earlier:

$y = A\cos 5x + B\sin 5x + \frac{3}{10}x\sin 5x$.

c

(d) Sketch $y = f(x)$ for $0 \le x \le \pi$. (2 marks)

d

Features

Exercise 7D — Core Pure Book 2 Pages 164–165 46

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Exercise 7D — Core Pure Book 2, pages 164–165.

Exercise 7D — Core Pure Book 2 Pages 164–165 48

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Exercise 7D — Core Pure Book 2 Pages 164–165 50

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